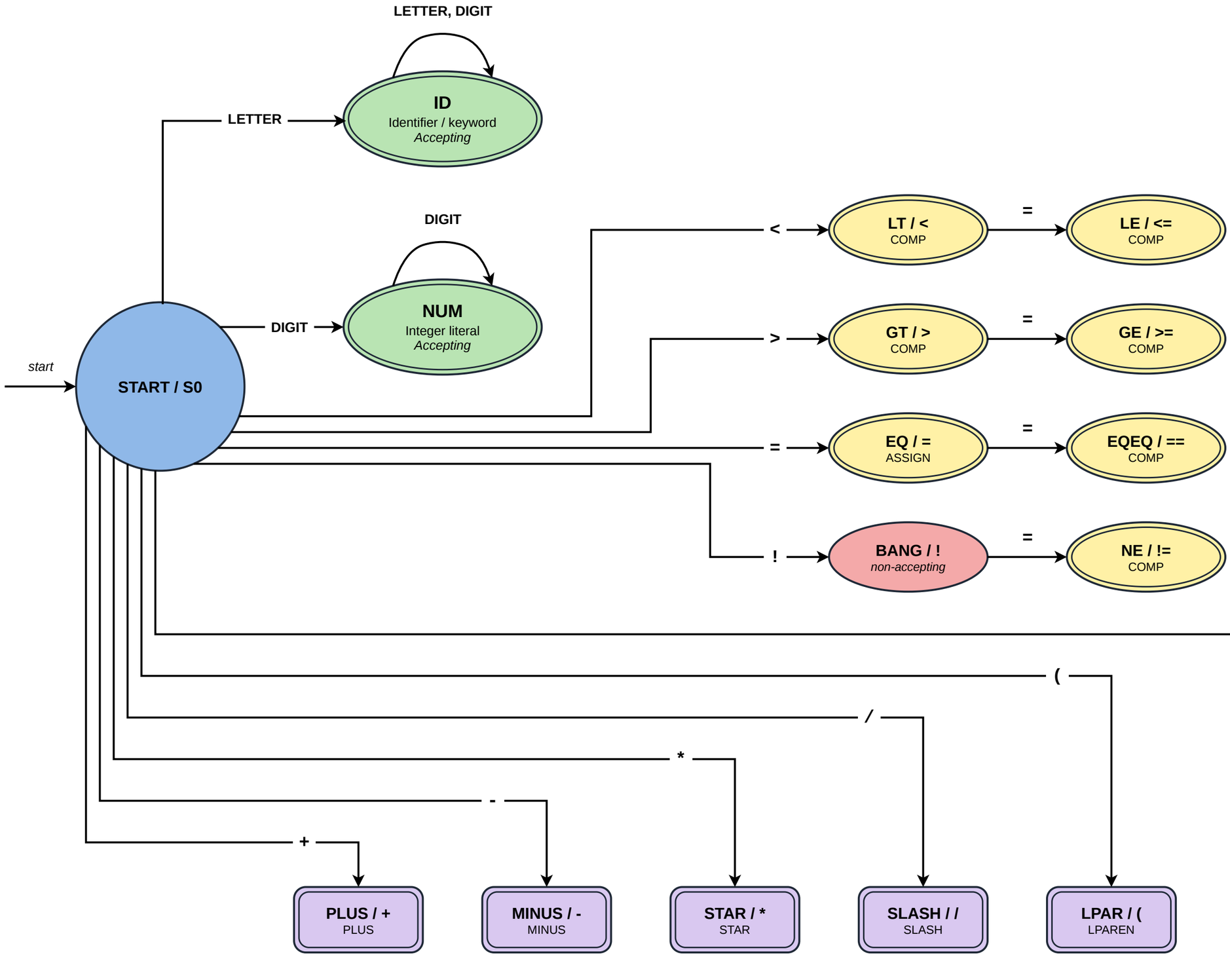
DFA-Based Tokenizer — Mini Programming Language

State transitions for identifiers, integer literals, and operators.

Input is scanned from left to right. The tokenizer uses maximal munch to recognize the longest valid token.



Legend

- Blue: Start state
- Green: Accepting identifier and integer states
- Yellow: Accepting operator states
- Red: Non-accepting intermediate state
- Purple: Accepting single-character token states

Implementation notes

- ID recognizes identifiers and keywords.
- NUM recognizes integer literals.
- LETTER = [A-Za-z_] DIGIT = [0-9]
- = produces ASSIGN.
- <, >, <=, >=, ==, != produce COMP.
- ! alone is invalid.
- Whitespace and newline characters are skipped.
- # comments continue to the end of the line.
- Unsupported characters raise LexError.
- The tokenizer uses maximal munch.
- Second line inside a state = token type emitted.